

Quantum Infrastructure Failure Testing Framework (QIFTF)

A Comprehensive Research and Analysis for the Design, Necessity, Applications, Funding, Ethics, and Red Team Analysis of a Purpose-Built Failure Condition Testing System for Quantum-AI Infrastructure

Executive Summary

Humanity is in the early stages of building a new layer of civilization-scale infrastructure: quantum computers, quantum-AI hybrid systems, and quantum communication networks. This infrastructure will, within the next decade, underpin global financial systems, national security architecture, pharmaceutical discovery, climate modeling, and the foundations of future artificial intelligence^{[1][2]}. Unlike any infrastructure built before it, quantum infrastructure does not simply run on electricity and software. It depends on physics itself — on the preservation of quantum states measured in microseconds, cryogenic environments operating near absolute zero, and probabilistic entanglement links that classical mathematics cannot fully replicate^{[3][4][5]}.

No rigorous, purpose-built framework exists today to stress-test the full dependency stack of this infrastructure against failure conditions. The frameworks that exist — chaos engineering for distributed software, digital twins for physical structures, and wargaming exercises for cyber incidents — were designed for classical systems and cannot capture the novel and fundamental failure modes of quantum infrastructure^{[6][7][8]}. This gap is not academic. It is a civilization-level risk.

The **Quantum Infrastructure Failure Testing Framework (QIFTF)** proposed in this report is a design, methodology, and policy blueprint for a systematic, multi-layer, multi-domain failure condition simulation and testing system for quantum-AI infrastructure. Its goal is distinct from

anything currently being pursued: not to make quantum systems perform better, but to discover how they break, how those breaks cascade, and whether society can survive those cascades.

Part I: What Quantum Infrastructure Actually Depends On

The Dependency Stack: Beneath the Qubit

Public discourse about quantum computing focuses on qubits, algorithms, and computational power. What is rarely discussed is the extraordinary physical and systemic architecture that qubits require to exist at all. Before a quantum computer can do a single calculation, the following dependencies must all be functioning simultaneously:

Layer 1 — Physical Environment (No Classical Equivalent)

Superconducting qubits — the dominant qubit type used by Google, IBM, and Rigetti — must be cooled to between 10 and 20 millikelvin (mK), temperatures colder than outer space^{[3][4]}. This requires dilution refrigerators drawing continuous power, and any interruption to cooling destroys the quantum state entirely. Trapped-ion qubits operate at approximately 4 Kelvin, still requiring laser cooling and ultra-high vacuum environments^{[3][5]}. Physical vibrations, electromagnetic radiation, and even cosmic rays can collapse qubit coherence^[^5]. Current quantum computers have error rates of approximately 0.1% to 1% per gate operation — meaning one out of every 100 to 1,000 operations introduces an error^[^9]. Achieving the fault-tolerant threshold requires reducing this to one error per million operations^[^10].

Layer 2 — Energy and Cooling Supply Chain

In classical computing, cooling represents 2–20% of total power consumption. In quantum computing, cryogenic cooling is the dominant power consumer — a complete inversion of the classical model^[^3]. Large-scale quantum systems require liquid helium and rare earth materials with highly constrained global supply chains^[^11]. Scaling to the thousands of logical

qubits required for cryptographically relevant computation will require cryogenic infrastructures of a scale never built before^{[12][13]}.

Layer 3 — Classical Control Electronics and Quantum-Classical Interface

Every quantum processor requires a classical control system that translates code into precisely timed microwave or laser pulses to manipulate qubits^[14]. This quantum-classical interface is arguably the most fragile integration point in the entire stack — and has no standardized testing protocol^{[12][14]}. Scaling fault-tolerant quantum computers will require heterogeneous quantum-classical architectures placing electronics at cryogenic stages, introducing new failure surfaces that have never been characterized at scale^[12].

Layer 4 — Timing, Synchronization, and Entanglement Distribution

Quantum networking requires that entangled photons be distributed across fiber optic links with sub-femtosecond timing precision^[15]. Entanglement distribution is inherently probabilistic — quality degrades with distance, and fidelity must be continuously maintained through entanglement purification and quantum repeater networks^{[16][17]}. A 10-user quantum network demonstrated at metropolitan scale maintained entanglement distribution over 10.8 days with an average secret key rate of only 3.38 bits per second — a far cry from the bandwidth required for civilization-scale quantum communication^[18].

Layer 5 — Quantum Error Correction and Software Stack

Even with physical hardware functioning, quantum error correction (QEC) must continuously operate to detect and correct errors without directly measuring the fragile logical qubit states^{[10][19]}. QEC requires enormous overhead: current estimates suggest that hundreds to thousands of physical qubits are required to produce a single reliable logical qubit^{[5][10]}. The software stack for decoding error syndromes must operate faster than errors accumulate — a real-time classical computation challenge that scales with system complexity^[20].

Layer 6 — Cryptographic Infrastructure and Post-Quantum Transition

All classical communication that supports quantum infrastructure — network management, software updates, authentication, key exchange — currently relies on cryptographic algorithms that a sufficiently powerful quantum computer will be able to break^{[21][22]}. NIST finalized post-quantum cryptographic (PQC) standards in August 2024, including ML-KEM, ML-DSA, and SLH-DSA^[23]. However, 91.4% of top websites still lack PQC support^[24], and a 2025 executive order requires federal systems to support PQC-ready protocols by 2030^[21]. The transition is urgent, expensive, and lagging behind the threat.

Layer 7 — Societal and Geopolitical Dependencies

Quantum infrastructure cannot function in societal isolation. It depends on rare material supply chains subject to geopolitical disruption, on an international scientific workforce with irreplaceable domain expertise, on regulatory frameworks that do not yet exist for many quantum applications, and on public trust that has never been discussed or solicited^{[25][26]}.

The Full Dependency Map

Layer	Key Dependency	Classical Analogue?	Primary Failure Mode
Physical environment	Cryogenic cooling, vibration isolation	No	Decoherence; thermal noise
Energy/supply chain	Ultra-stable power; liquid helium	Partial	Cooling failure; material scarcity
Control electronics	Quantum-classical interface hardware	Partial	Signal timing errors; crosstalk
Networking	Fiber; quantum repeaters; timing sync	No	Entanglement degradation; latency
Error correction	QEC software; decoder algorithms	No	Error rate exceeds fault tolerance threshold
Cryptography	PQC-ready protocols, PKI migration	Partial	HNDL attacks; cryptographic legacy debt

Societal	Supply chains; governance; expertise	No	Geopolitical disruption; regulatory failure
----------	---	----	--

Part II: Why This Framework Is Needed — The Affirmative Case

The Infrastructure Gap No One Is Testing

The history of catastrophic infrastructure failure is consistently a history of untested interdependencies. The 2003 Northeast Blackout affected 55 million people across eight U.S. states and two Canadian provinces — originating from a single software bug in an alarm system that caused a cascade no one had modeled^{[27][28]}. Hurricane Maria exposed cascading interdependencies in Puerto Rico's power, water, communications, and healthcare systems that had never been analyzed as a combined system^{[29][30]}. These were classical systems, with decades of operational experience, and they failed in ways no one had tested.

Quantum infrastructure is orders of magnitude more complex, more fragile, and more interdependent than anything these historical failures involved. And it is being deployed against a timeline measured in years, not decades.

The "Harvest Now, Decrypt Later" (HNDL) Problem Is Already Active

The HNDL threat model is not theoretical. Nation-state actors are currently collecting encrypted data with the explicit intent of decrypting it once quantum computers mature^{[31][32]} [^33]. A 2026 Federal Reserve analysis documents how distributed ledger cryptocurrency networks face permanent exposure through HNDL, because previously recorded transactions cannot be retroactively protected after the fact[^34]. Data captured in 2026 could be decrypted in 2032 — a breach that is invisible today and catastrophic later[^31]. A failure testing framework for quantum infrastructure must treat this as an active, live failure condition, not a future scenario.

Cascading Failure Risk Is Systematically Underanalyzed

Critical infrastructure interdependency analysis from Argonne National Laboratory demonstrates that a threat or hazard can create cascading failures with a "multiplicative effect on risk" — impacts propagating across sectors in ways that "mask many systemic risks" and can produce "significant economic and physical damage on a city-wide, regional, national, or international scale"[²⁷]. Applied to quantum infrastructure: a failure in the cryogenic helium supply chain does not just affect quantum computers. It affects quantum key distribution networks, which affects the security of financial transactions, which affects market confidence, which can produce economic cascades far disproportionate to the initial physical disruption.

No Existing Testing Framework Addresses This

The Quantum Internet Research Group (QIRG) published architectural principles in RFC 9340 (2023), but explicitly states that "there is no practical proposal for how to organise, utilise, and manage such networks" and that it is "intended for general guidance"[³⁵][³⁶]. The EU's Quantum Internet Framework Partnerships program (HORIZON 2025) targets entanglement distribution over 500 km and quantum repeater integration — but its success criteria are performance benchmarks, not failure condition characterization[³⁷]. QAISim, a 2024 toolkit for modeling AI in quantum cloud environments, focuses exclusively on task scheduling and resource allocation[³⁸]. None of these constitute a failure condition testing framework.

Chaos Engineering — the most rigorous existing paradigm for failure injection — was explicitly designed for classical distributed software systems. Netflix's Chaos Monkey terminates virtual machine instances and microservices[⁶][³⁹][⁸]. It has no conception of decoherence, cryogenic failure, entanglement degradation, or quantum-classical interface instability. Applying it unchanged to quantum infrastructure would produce meaningless results.

Why This Matters Now, Not Later

Two independent data points establish the urgency:

First, Google's Willow processor performs calculations in under five minutes that would require the Frontier supercomputer ten septillion years[¹]. IBM's Quantum System Two, with

156 qubits, operates 50 times faster than its predecessors^[1]. The transition from laboratory curiosity to civilization-altering tool is not decades away — it is underway.

Second, quantum infrastructure failure does not look like a server going offline. Decoherence is instantaneous and invisible. Entanglement degradation produces incorrect results, not error messages. A failure in quantum error correction produces outputs that look correct but are not — the most dangerous failure mode in any safety-critical system. Understanding how these failures propagate before they happen in deployed systems is not just prudent. It is the only responsible path forward.

Part III: Framework Design — The QIFTF

Guiding Philosophy

The QIFTF is built on a single organizing principle that distinguishes it from all existing frameworks: **test the physics, not just the software**. This means:

- Failure injection must extend below the software layer into the physical and environmental substrates that quantum systems depend on
- Success criteria must be defined in terms of societal continuity, not system uptime
- Testing must be cross-domain — involving quantum physicists, network engineers, supply chain analysts, cryptographers, emergency management professionals, and policymakers simultaneously
- Every simulation must include an explicit "cannot recover" failure pathway — the most important test is whether civilizational dependencies have been maintained alongside quantum capability, or whether they have been eliminated in favor of quantum efficiency

Framework Architecture: Five Interlocking Pillars

Pillar 1 — Physical Substrate Failure Injection

This pillar tests what no existing framework has ever tested: what happens when the physics-level dependencies of quantum systems fail.

Test scenarios include:

- Controlled cryogenic cooling degradation: staged temperature increases from 15 mK toward operating-failure threshold, measuring at what point error rates exceed QEC capacity and computation becomes unreliable
- Electromagnetic pulse (EMP) simulation: application of controlled EM fields to test shielding adequacy and recovery time for quantum-classical control electronics
- Vibration injection: introduction of seismic or mechanical vibration at different frequencies to characterize decoherence susceptibility and establish safe deployment envelope for quantum hardware in non-laboratory environments
- Power fluctuation profiles: introduction of realistic power grid instability patterns (simulating grid stress, renewable intermittency, surge events) to test system response and identify the minimum power stability specification for continuous operation
- Helium supply disruption simulation: model the operational response to liquid helium supply reduction of 10%, 25%, 50%, and 100% over different timeframes

Pillar 2 — Network and Entanglement Failure Injection

This pillar tests the quantum communication infrastructure, where failure modes have no classical parallel.

Test scenarios include:

- Fiber cut injection in entanglement distribution networks: what is the minimum network topology for maintaining entanglement distribution under N-1 and N-2 link failures?
- Timing synchronization disruption: introduce clock drift in the sub-femtosecond range to characterize the degradation curve of entanglement fidelity as synchronization degrades

- Quantum repeater failure: test entanglement routing under progressive repeater node failures; establish maximum tolerable node loss before end-to-end quantum communication fails
- Entanglement fidelity degradation cascade: at what fidelity threshold does a quantum network's output become indistinguishable from noise, and how quickly does that threshold spread through a multi-node network?
- Satellite quantum link failure: simulate atmospheric disturbance, orbital coverage gaps, and ground station failures for space-based quantum key distribution

Pillar 3 — AI and Classical Control Failure Injection

This pillar tests the quantum-classical interface and AI-in-the-loop components — the connective tissue between quantum hardware and human-usable systems.

Test scenarios include:

- Quantum error correction decoder failure: introduce deliberate decoder errors to characterize how quickly incorrect classical error estimates propagate into catastrophic logical qubit failure
- AI-quantum hybrid system misalignment: inject adversarial inputs into the AI components of hybrid systems and observe whether the quantum components amplify or absorb the error
- Real-time simulation overload: stress-test quantum-in-the-loop (QIL) systems under conditions of extreme sensor density and update frequency, consistent with NREL's smart grid QIL framework^[40][41]
- QAI data risk injection: operationalize the 22 data risk categories identified in the 2025 taxonomy of QAI risks, including governance failures, control implementation gaps, and user-level vulnerabilities^[42]
- Classical fallback validation: for every quantum-dependent function, test that a viable classical fallback exists, can be activated within a defined latency window, and produces results within acceptable degradation parameters

Pillar 4 — Cryptographic and Security Failure Injection

This pillar stress-tests the cryptographic transition from classical to post-quantum standards, including the active HNDL threat.

Test scenarios include:

- HNDL simulation: in a controlled environment, simulate a 5-year HNDL attack against a quantum-dependent network; quantify what data categories remain exposed and model the societal impact of their decryption
- PQC migration failure: simulate a forced cutover to NIST-standardized PQC algorithms (ML-KEM, ML-DSA, SLH-DSA) under adversarial conditions, testing interoperability failures, latency increases, and performance degradation against real-world baselines
- Cryptographic inventory gap: model the consequences of a PQC transition attempted without a prior cryptographic inventory — the scenario that currently describes the majority of global organizations^{[43][44]}
- Hybrid classical-quantum cryptography failure: test the security of hybrid systems (combining classical and PQC algorithms) under adversarial quantum decryption capability
- Crypto-agility validation: stress-test whether systems can switch cryptographic algorithms mid-operation in response to a discovered vulnerability — the capability known as crypto-agility^[45]

Pillar 5 — Societal and Cascade Failure Simulation

This is the pillar that makes QIFTF categorically different from any technology testing framework in existence. It tests not whether the quantum infrastructure survives a failure, but whether society survives the quantum infrastructure failing.

Test scenarios include:

- Power grid + quantum infrastructure simultaneous failure: using a Quantum-in-the-Loop real-time simulation approach, model a scenario where quantum-dependent power grid

optimization systems fail due to decoherence while the classical grid is in a degraded state; measure the cascade depth

- Financial system quantum dependency failure: in collaboration with financial regulators, model a scenario where quantum-dependent fraud detection, clearing, and settlement systems simultaneously fail; measure the systemic financial impact
- Healthcare quantum dependency failure: for hospitals and research institutions that have integrated quantum computing into drug discovery pipelines, imaging analysis, or radiation treatment planning, model failure scenarios
- Cross-sector cascade wargame: run a tabletop exercise using the "resilience games" framework developed by the U.S. Army Cyber Institute's Jack Voltaic series^[46], adapted for quantum infrastructure failure scenarios across energy, finance, defense, healthcare, and communication sectors
- Civilization floor test: the most important test in the entire framework — for each critical societal function currently being migrated to quantum infrastructure, can it be performed, even at degraded capacity, using pre-quantum technology? This tests whether classical redundancy has been maintained or eliminated

Testing Protocol Structure

Phase	Name	Duration	Method	Output
Phase 1	Dependency Mapping	12–18 months	System audit; supply chain analysis; interdependency graph construction	Comprehensive QITF dependency atlas
Phase 2	Physical Failure Characterization	24–36 months	Laboratory-controlled failure injection in isolated quantum hardware testbeds	Physical failure envelope specifications per qubit type and system configuration

Phase 3	Network Failure Simulation	18–24 months	Simulation against deployed testbeds (e.g., Argonne ArQNet ^[47] , EU Quantum Internet)	Network resilience specifications; minimum viable topology definitions
Phase 4	Cryptographic Transition Stress Testing	12–18 months	Red team adversarial testing of PQC migration; HNDL simulation	Cryptographic vulnerability registry; transition risk quantification
Phase 5	Societal Cascade Simulation	24–36 months	Wargaming; digital twin modeling; cross-sector tabletop exercises	Civilization floor assessment; cascade propagation maps
Phase 6	Integrated Full-Stack Failure Exercise	12 months (annual)	Combined simulation across all five pillars with live participants	Annual QITF State of Quantum Resilience Report

Part IV: How This Is Fundamentally Different from Everything Being Built Today

The Civilizational Gap in Current Quantum Development

The world's major quantum programs — DARPA, the EU Quantum Flagship, China's National Quantum Laboratory, IBM's Quantum Network, Google's Quantum AI division — are all united by a single organizing goal: make quantum systems work better. Their metrics are qubit count, gate fidelity, error rates, entanglement distance, and algorithmic performance^{[1][2][37]}. Every dollar spent, every paper published, every grant awarded is directed toward making quantum infrastructure more powerful.

This is correct and necessary. It is also profoundly incomplete.

No major program — national, academic, or private — is primarily organized around the question: *what happens when this infrastructure fails?* More specifically: *what happens when this infrastructure fails and humanity has already come to depend on it?*

The QIFTF is not a better quantum computer program. It is not a post-quantum cryptography program. It is not even an infrastructure resilience program in the traditional sense. It is a **civilizational risk assessment program** organized around a simple, unprecedented, and deeply uncomfortable question: are we prepared for the world we are building?

Why Existing Paradigms Are Categorically Insufficient

Chaos Engineering (Netflix/Gremlin model): Chaos Engineering was built on the premise that failure in distributed systems is inevitable, random, and primarily software-defined^{[6][48]}^[^8]. It injects failures into running systems and observes whether they recover gracefully. This is powerful for classical microservices, but it cannot address: (1) physics-level failures with no software analogue; (2) failure modes whose output looks correct but is not; (3) cascades that emerge from the interdependency between quantum and classical systems rather than within either alone; or (4) failures whose consequences materialize years after the initial breach (HNDL).

Digital Twins: Digital twins create virtual replicas of physical systems to enable simulation without disrupting operations^{[7][49]}^[^50]. For quantum systems, this approach faces a fundamental physical limit: simulating a quantum system on a classical computer requires exponentially more computational resources as the quantum system grows. A 50-qubit quantum system would require a classical simulation of more than a quadrillion simultaneous states^[^51]. You cannot build a faithful digital twin of a large quantum system on classical hardware. The QIFTF must therefore use actual quantum hardware testbeds for physical failure injection, not simulation substitutes.

Infrastructure Resilience Wargaming: The Jack Voltaic series and "preparedness wargaming" approaches are the most conceptually adjacent paradigms to QIFTF^[46]^[52]. They model cascading infrastructure failures across sectors and generate policy recommendations. However, they are designed by and for classical infrastructure operators. Their failure

scenarios are based on cyberattacks, natural disasters, and supply chain disruptions — all of which are relevant but none of which includes the physics-layer failures, decoherence-induced silent errors, or HNDL-class long-horizon vulnerabilities that quantum infrastructure introduces.

NIST/PQC Standards Programs: NIST's post-quantum cryptographic standardization effort is the most significant quantum-related policy initiative underway^{[21][23]}. It is technically rigorous and organizationally impressive. But it addresses one dependency layer — cryptography — and explicitly does not address the physical hardware, network architecture, supply chain, AI integration, or societal cascade dimensions of quantum infrastructure failure.

What QITF Does That No Existing Framework Addresses:

Dimension	Chaos Engineering	Digital Twins	Wargaming	NIST/PQC	QITF
Physical-layer failure injection	No	Partially	No	No	Yes
Decoherence-specific testing	No	No	No	No	Yes
Quantum-classical interface failure	No	Partially	No	No	Yes
Entanglement network failure	No	No	No	No	Yes
Supply chain disruption	No	No	Partially	No	Yes
HNDL-class threat simulation	No	No	Partially	Partially	Yes

Societal cascade simulation	No	No	Partially	No	Yes
Civilization floor test	No	No	No	No	Yes
Cross-domain, cross-sector scope	No	No	Partially	No	Yes
Physics-faithful simulation	N/A	No	N/A	N/A	Yes

Part V: Funding Landscape

Why Quantum Infrastructure Resilience is Underfunded

The global investment in quantum computing development is substantial and accelerating. McKinsey's 2025 Quantum Technology Monitor documents the transition "from concept to reality" across computing, sensing, and communication domains^[^2]. The EU's HORIZON quantum programs are deploying hundreds of millions of euros toward building the quantum internet^[^37]. The U.S. federal government has mandated post-quantum cryptography migration through multiple executive orders and national security memoranda^[^{21]}44].

What is not being funded — anywhere, at scale — is the failure condition side of the equation. This mirrors a pattern familiar from other technology transitions: enormous investment in capability development, neglect of resilience, followed by costly and preventable failures when the technology scales.

Recommended Funding Pathways

Federal Government (U.S.)

- DARPA: Propose QITF under the Information Innovation Office (I2O) as a novel program addressing national security implications of quantum infrastructure failure; precedent includes the acoustic warfare payload contracts (\$20M+ cumulative)[⁵³] and force-detecting sensor grants (\$2M, Penn State)[⁵⁴]
- DHS/CISA: Critical infrastructure protection mandate covers 16 sectors that quantum infrastructure will affect; QITF aligns directly with CISA's resilience and risk assessment mission
- NIST: Natural institutional home for physical failure characterization and standard-setting for quantum hardware failure modes; existing PQC work creates an administrative pathway
- DOE: National Laboratories (Argonne, NREL, Oak Ridge) are already active in quantum network testbeds[⁴⁰][⁴¹][⁴⁷]; QITF can leverage this existing infrastructure
- NSF: Basic research funding for the interdisciplinary methodology development — quantum physics, network science, resilience engineering, and social science integration
- Estimated Phase 1–2 federal funding: \$20,000,000–\$100,000,000 over 5 years

International Programs

- EU Quantum Flagship: HORIZON-CL4 programs already funding quantum internet infrastructure[³⁷]; a QITF component can be proposed under the "resilience and security" objectives
- NATO: Quantum infrastructure failure scenarios have direct defense implications; NATO's emerging technology programs are a natural funding mechanism
- OECD / G7: The G7 Cyber Expert Group's January 2026 roadmap for financial sector PQC transition[⁵⁵] establishes the institutional framework for multi-lateral quantum resilience funding

Private Sector

- Quantum hardware vendors (IBM, Google, IonQ, Quantinuum): have a commercial interest in validated failure characterization — it is a prerequisite for enterprise customer adoption at scale; could fund Pillar 1 and Pillar 3 components
- Insurance and reinsurance industry: quantum infrastructure failure represents an emerging class of systemic risk for which insurers currently have no actuarial model; funding QIFTF produces the loss models they need
- Financial services: given direct exposure to HNDL and quantum-dependent market infrastructure, major banks and clearinghouses have institutional interest in Pillar 4 funding^[55][34]

Intellectual Property and Revenue Model

- QIFTF can generate licensing revenue through the certification standard — organizations whose quantum infrastructure has been tested against QIFTF protocols can receive a certification analogous to ISO 27001 for information security, creating a market-based revenue stream that reduces long-term dependence on grant funding

Part VI: Ethical Implications

Power Asymmetry and Quantum Stratification

Quantum computing capability will not be distributed evenly. Nations, organizations, and social groups that achieve quantum advantage — in computational power, cryptographic security, and AI enhancement — will hold structural advantages over those that do not^[56][26]. A failure testing framework that is proprietary, classified, or accessible only to wealthy nations or corporations could itself become a tool of stratification: those with QIFTF certification will be trusted; those without will not. The framework must be designed from the outset with equity principles embedded — not retrofitted when stratification becomes visible.

Dual-Use Risk of Failure Knowledge

Every failure mode characterized by QIFTF is potentially exploitable. Understanding precisely at what point cryogenic cooling disruption causes quantum error correction to fail is useful for designing resilient systems — and for adversaries seeking to disable those systems. This is the dual-use dilemma in its starkest form^{[57][58]}. The framework must establish a tiered information release protocol — analogous to responsible disclosure in cybersecurity — that separates publicly available methodological knowledge from operationally sensitive vulnerability characterizations.

Governance Without a Legal Framework

The UN declared 2025 the International Year of Quantum Science and Technology, acknowledging the "urgent" need to develop legal and regulatory frameworks for quantum technology^[^25]. As of 2026, those frameworks remain nascent. The QIFTF would generate findings — about quantum infrastructure vulnerabilities, about failure cascade pathways, about HNDL exposure in specific sectors — that have profound legal, regulatory, and national security implications, but no established international governance structure to manage them^[^26]. Establishing the institutional governance of QIFTF before its outputs become operational is not a bureaucratic formality. It is a prerequisite for responsible deployment.

Environmental and Resource Ethics

Quantum infrastructure's environmental footprint is dominated by cryogenic energy consumption — a demand profile radically different from classical computing, where energy efficiency has been a decades-long engineering focus^[^11]. QIFTF stress-testing must include environmental failure scenarios: what happens to a quantum infrastructure node when regional electricity grids are stressed by climate events? What happens when the rare earth and helium supply chains that quantum hardware depends on face geopolitical or climate disruption? The intersection of quantum infrastructure failure and climate vulnerability is entirely uncharacterized — and ethically urgent.

The Right to Know and Consent

Classical citizens have no meaningful understanding of the quantum infrastructure being built to underpin their financial systems, healthcare, and communications. Unlike previous technology transitions — the internet, mobile networks, cloud computing — quantum infrastructure introduces physics-level risks that are genuinely invisible to non-specialists. Decoherence is not a concept the public has a mental model for. HNDL attacks are currently ongoing and the people whose data is being harvested have no knowledge of this^{[32][33]}. An ethical QIFTF program must include a public communication component that translates its findings into terms that enable meaningful democratic oversight — not as a public relations exercise, but as a prerequisite for legitimate governance.

Part VII: Red Team Analysis — The Case Against

This section presents the strongest available arguments against the QIFTF as proposed. These are not the author's position but are included as intellectually honest counter-positions that any serious proposal must engage.

Technical Counter-Arguments

1. The Framework Cannot Be Faithful to the Physics It Seeks to Test

Simulating quantum failure classically is impossible at scale — a quantum system of 50 qubits requires more than a quadrillion simultaneous states to simulate faithfully^[^51]. This means QIFTF must use real quantum hardware testbeds for physical failure injection. But injecting failures into real quantum hardware risks damaging irreplaceable, extremely expensive, and technically irreplacable experimental systems. The framework faces a paradox: the most important tests require risking the hardware they are designed to protect.

2. Quantum Infrastructure Failure Modes Are Not Yet Fully Characterized

You cannot systematically test for what you cannot model. Decoherence research is still active and producing new findings about correlated errors, non-Markovian noise structures, and burst events that current QEC frameworks do not adequately address^[^20]. A QIFTF built today

would be testing against an incomplete taxonomy of failure modes. Fraunhofer's 2025 study on using quantum computers for resilience analysis of critical infrastructure found that "quantum computers with much lower error rates than currently available are required for quantitative resilience analysis"^[59]. The tools required to run the framework may not yet exist.

3. Individual Components Are Already Tested — Integration May Not Add Proportional Value

IBM tests its quantum hardware exhaustively before deployment. Argonne tests its ArQNet quantum network testbed^[47]. NIST tests its PQC algorithms against classical and quantum adversaries^{[23][60]}. Google tests its Willow processor for gate fidelity and error rates^[1]. The argument that an integrated QIFTF adds disproportionate value over this existing testing regime requires evidence that integrated failure modes produce emergent risks not visible in component testing. That evidence does not yet exist in the peer-reviewed literature.

4. Physical Layer Testing Could Produce Misleading Confidence

Goodhart's Law states that when a measure becomes a target, it ceases to be a good measure. A QIFTF certification could create false confidence — organizations and governments that have passed the framework's tests may assume their quantum infrastructure is safe, when in reality the framework's test coverage is inevitably incomplete. The history of financial stress testing is instructive: banks that passed pre-2008 stress tests still failed catastrophically, in part because the stress tests did not model the scenarios that actually occurred.

Strategic and Policy Counter-Arguments

5. This Is Premature — Quantum Infrastructure Isn't Deployed at Scale Yet

The CRQC (cryptographically relevant quantum computer) capable of breaking RSA encryption is projected for the 2030s at the earliest^{[21][43]}. Deployed quantum communication networks currently achieve secret key rates measured in bits per second^[18]. Civilization-scale quantum infrastructure dependence is likely a decade or more away. Spending significant resources now on a failure testing framework for infrastructure that doesn't exist at scale may

be premature — and could consume resources better directed at building the capabilities that will eventually need testing.

6. International Coordination Is Required but Geopolitically Impossible

Meaningful quantum infrastructure failure testing requires shared access to quantum hardware, network testbeds, and classified failure characterization data across national boundaries. In the current geopolitical environment — with intense U.S.-China competition in quantum technology, EU strategic autonomy initiatives, and the absence of any international quantum governance framework^{[25][26]} — achieving the multilateral cooperation QIFTF requires may be politically impossible. A framework that can only be run within national boundaries will have gaps that map precisely to the most dangerous failure vectors: cross-border quantum networks and internationally distributed supply chains.

7. Existing Resilience Frameworks Can Be Adapted More Efficiently

The investment required to build QIFTF from scratch — new testbeds, new interdisciplinary teams, new governance structures, new certification protocols — may be greater than the investment required to adapt existing frameworks. Chaos Engineering tools could be extended with quantum-specific failure injection plugins. NIST could be tasked with expanding PQC transition standards to cover hardware failure modes. Digital twin vendors could be required to include quantum emulation layers in next-generation products. The marginal cost of adapting existing paradigms may be substantially lower than the fixed cost of building a new one.

8. The "Civilization Floor" Test May Be Politically Untenable

The civilization floor test — verifying that every quantum-dependent critical function can still be performed using pre-quantum technology — directly challenges the economic logic driving quantum infrastructure adoption. Organizations adopting quantum technology are doing so, in part, to decommission expensive classical systems and eliminate redundancy costs. Requiring them to maintain classical fallback capacity indefinitely imposes substantial

ongoing costs that will face fierce political opposition from exactly the industries and governments that would need to fund QIFTF.

Ethical Counter-Arguments

9. The Framework Could Accelerate the Adversarial Exploitation It Seeks to Prevent

Characterizing quantum infrastructure failure modes in detail creates a roadmap for adversaries. Even with tiered information release protocols, information about specific physical failure thresholds, critical network topology vulnerabilities, and HNDL-exploitable data categories will inevitably be sought by state and non-state adversaries. The more comprehensive the framework, the more valuable the target. This is not an argument against security research — it is an argument for taking the security of the framework itself as seriously as the security of the infrastructure it tests.

10. Framing Quantum Risk Around Testing Could Displace Necessary Governance

There is a risk that QIFTF becomes a technical substitute for political accountability. Testing frameworks can be used — and historically have been used — to create the appearance of governance without the substance. If governments and corporations point to QIFTF certification as evidence of quantum safety, while simultaneously resisting the binding international agreements, export controls, and legal frameworks that quantum governance actually requires^{[58][25][^26]}, the framework will have made things worse by providing cover for inaction on harder political problems.

Part VIII: Why the Goal Must Be Fundamentally Different

A Mission Statement That Does Not Yet Exist

Every quantum program in the world today is a performance program. Its mission is to make quantum systems do more, faster, with higher fidelity. This is the correct mission for the organizations running those programs — IBM, Google, the EU Quantum Flagship, China's

national laboratories, DARPA. They are building a capability. They should be building it as well and as fast as possible.

But civilization does not need only capability. It needs survivability. It needs the answer to a question that no capability program can provide: if this infrastructure fails — partially, completely, gradually, or catastrophically — what happens to the people who depend on it?

QIFTF's mission is not to make quantum infrastructure better. Its mission is to determine whether humanity is ready to depend on quantum infrastructure at all — and to define the conditions under which that dependence becomes safe.

This is a fundamentally different category of program. It is not a research program, an engineering program, or a policy program. It is a **civilizational preparedness program** — a category that does not yet exist in the institutional landscape of quantum technology development.

The Analogies That Illuminate the Gap

The closest historical analogues are:

- The creation of nuclear safety and nonproliferation regimes *alongside* the deployment of nuclear power and weapons — not as an afterthought, but as a parallel institutional development
- The establishment of aviation safety certification standards by the FAA — which does not build aircraft but determines the conditions under which aircraft can carry people
- The development of biosafety level (BSL) protocols for pathogen research — which do not advance microbiology but determine the conditions under which dangerous biological research can be conducted

In each case, the safety institution was not part of the capability-building enterprise. It was separate from it, empowered to constrain it, and measured by a different success criterion: not how much capability was developed, but how safely it could be deployed.

QIFTF occupies exactly this position relative to the global quantum technology enterprise. Its success is not measured in qubit counts or gate fidelities. It is measured in the answer to one question: *when the quantum era arrives, is humanity prepared for it?*

Conclusion

The Quantum Infrastructure Failure Testing Framework is not a proposal to slow quantum development. It is a proposal to make quantum development survivable. The physical dependencies of quantum infrastructure — cryogenic cooling, vibration isolation, rare material supply chains, probabilistic entanglement, quantum error correction, and a cryptographic transition already underway under adversarial conditions — represent a failure risk profile with no precedent in the history of infrastructure development^{[3][5][15][1]}.

The strongest arguments in QIFTF's favor are the active HNDL threat (a breach already in progress), the demonstrated history of cascading infrastructure failures in systems far simpler than quantum infrastructure^{[29][30][^28]}, the categorical inadequacy of existing testing paradigms for the physics-layer failure modes quantum systems introduce, and the irreversibility of civilization-level dependence once established.

The strongest arguments against are the immaturity of quantum systems (the most important tests may require hardware that does not yet exist at scale), the dual-use risk of failure characterization, the political difficulty of requiring classical fallback capability maintenance, and the risk that certification creates false confidence.

A responsible development program would proceed immediately with Phase 1 (dependency mapping) and Phase 4 (cryptographic stress testing), as these require no quantum hardware and address active threats. Physical failure injection (Phase 2) should begin in parallel at the most advanced quantum testbed facilities, within explicit experimental risk parameters. The civilization floor test (Phase 5) should be treated as a political as well as a technical project — requiring engagement from elected officials, not just engineers.

The quantum era is arriving whether humanity is prepared for it or not. QITF is the proposal that preparation is possible, necessary, and — if begun now — still achievable.

Report prepared April 2026. All cited data sourced from peer-reviewed research publications, government agency documentation, national laboratory research, regulatory filings, and international standards bodies.

References

1. [The Quantum Era Is Upon Us - Forbes](#) - The advancements achieved in 2025–2026, such as scalable logical qubits, clear benefits of quantum c...
2. [The Year of Quantum: From concept to reality in 2025 - McKinsey](#) - Explore the latest advancements in quantum computing, sensing, and communication with our comprehens...
3. [Designing Energy-Efficient Quantum Computers Through Prediction and Reduction of Cooling Requirements for Cryogenic Electronics](#)
4. [What Is Cryogenic Quantum Computing and Why It Matters - SpinQ](#) - Discover how cryogenic cooling powers superconducting qubits. Learn why ultra-low temperatures are e...
5. [What Is Quantum Error Correction & How Does It Work](#) - Learn how quantum error correction works, why it matters, and which companies are leading the race t...
6. [What is Netflix's Chaos Monkey? - GeeksforGeeks](#) - Your All-in-One Learning Portal: GeeksforGeeks is a comprehensive educational platform that empowers...
7. [Why Traditional Safety Testing Fails; What We're Building Instead - ... 2025](#), by integrating fault scenarios and enabling simulation “dress rehearsals” to surface proc...

8. [Chaos Engineering](#) - Modern software-based services are implemented as distributed systems with complex behavior and fail...
9. [Quantum error correction](#) - Details methods like surface codes and stabilizer codes used to protect information against errors i...
10. [Why do quantum computers need QEC - Riverlane](#) - Today's quantum computers have high error rates – around one error in every few hundred operations. ...
11. [Sustainable generative AI and quantum computing - Frontiers](#) - It first examines the scope of what is measured in life cycle assessments, then analyzes the timing ...
12. [Integration and Resource Estimation of Cryoelectronics for ... - arXiv](#)
13. [Thermal capacity mapping of cryogenic platforms for quantum ... - OQC](#)
14. [What is quantum-classical integration](#) - Essentially, the quantum control system acts as a translator. It converts classical code into precis...
15. [From Secure Links to Entangled Systems: A Roadmap for Global ... - By distributing](#) entanglement across fiber, free-space, and satellite channels, quantum networks prov...
16. [Quantum Networking 101: Using Entanglement - Aliro Technologies](#) - The distributed entanglement must have high quality. · The network needs high enough throughput, or ...
17. [Optimal routing and end-to-end entanglement distribution in ... - Nature](#) - This article investigates the engineering problem of resource allocation in quantum networks, consid...
18. [\[2211.09051\] Entanglement distribution quantum networking within ...](#) - We show a 10 user fully connected quantum network with metropolitan scale deployed fibre links, demo...
19. [\[2202.08600\] Decoherence and Quantum Error Correction ...](#) - Quantum technologies have shown immeasurable potential to effectively solve several information proc...
20. [File 1 - OPEN PEER REVIEW](#)

21. [Building Resilient Security in the Age of Quantum Computing - ISACA](#) - In April 2024, the European Commission published a recommendation urging member states to adopt a ha...
22. [\[PDF\] Post-Quantum Financial Infrastructure Framework \(PQFIF\) - SEC.gov](#) - 2.2.1 Quantum Computing Landscape Evolution (2024-2025). Recent Quantum Computing Developments: Rece...
23. [Industrial systems face structural gap as quantum risks drive ...](#) - ... 2024, with quantum-vulnerable algorithms targeted for complete transition by 2035. Though high-r...
24. [The Post-Quantum Cryptography Crisis: Why 2026 Will Be a ...](#) - 91.4% of top websites lack post-quantum cryptography support, while quantum computers threaten curre...
25. [International Dialogue on Quantum Legal Frontiers 2025](#) - The unique principles of quantum mechanics challenge traditional notions of data security, intellect...
26. [\[PDF\] Global Quantum Governance: From Principles to Practice](#) - Much like AI, quantum technology puts pressure on regulatory systems that were built around slower i...
27. [\[PDF\] Analysis of Critical Infrastructure Dependencies and](#) - These strategic directives reveal the importance of analyzing infrastructure dependencies, interdepe...
28. [The Danger of Critical Infrastructure Interdependency](#) - This essay will discuss a method to assess the risks to critical infrastructure that result from int...
29. [\[PDF\] Critical Infrastructure Interdependency Analysis - PreventionWeb.net](#) - In this phase, the assessment team analyses the data collected for characterising critical assets' d...
30. [Critical infrastructure interdependency analysis: Operationalising ...](#) - This paper proposes a critical infrastructure interdependency analysis framework and illustrates its...

31. ["Harvest Now, Decrypt Later" Quantum Threat for Regulated Industries](#) - HNDL changes the timeline of cyber risk. A breach enabled by quantum computing in 2032 may originate...
32. [Harvest Now, Decrypt Later \(HNDL\): The Quantum-Era Threat](#) - HNDL is a cybersecurity threat where encrypted data is collected and stored so it can be decrypted w...
33. [Quantum cryptography and the "harvest now, decrypt later" problem](#) - The threat isn't theoretical anymore. Nation-state actors are already believed to be collecting encr...
34. [The Fed - "Harvest Now Decrypt Later": Examining Post-Quantum ...](#) - This paper analyzes the risks posed by future-state quantum computers, specifically the "harvest now...
35. [Architecture of a Quantum...](#) - The vision of a quantum internet is to enhance existing Internet technology by enabling quantum comm...
36. [RFC 9340 - Architectural Principles for a Quantum Internet](#) - The vision of a quantum internet is to enhance existing Internet technology by enabling quantum comm...
37. [Quantum Internet Framework Partnerships Agreement ...](#)
38. [QAISim: A Toolkit for Modeling and Simulation of AI in Quantum ...](#)
39. [Chaos Monkey Guide for Engineers - Gremlin](#) - A complete and comprehensive guide to learn about, set up, and deploy Chaos Monkey and other similar...
40. [Architecture for Quantum-in-the Loop Real-Time Simulations for Designing Resilient Smart Grids](#)
41. [Architecture for Quantum-in-the Loop Real-Time Simulations for ...](#)
42. [A Taxonomy of Data Risks in AI and Quantum Computing \(QAI\) - arXiv](#) - However, QAI inherits data risks from both AI and QC, creating complex privacy and security vulnerab...

43. [Secure Today, Vulnerable Tomorrow? Quantum Computing and the ...](#) - While you may think your encryption is future proof, advancements in quantum computing suggest what ...
44. [Year in Review 2025: Insights on Post-Quantum Readiness](#) - A 2025 retrospective on post-quantum readiness — milestones, lessons, and strategies shaping the tra...
45. [Challenges & Adoption of Post-Quantum Cryptography](#) - Faced with the latent cyber threat, post-quantum cryptography (PQC) is becoming a priority for busin...
46. [\[PDF\] Introducing Resilience Games for Critical Infrastructure Preparedness](#)
47. [Experimental Demonstration of Software-Orchestrated Quantum ...](#)
48. [Chaos Monkey at Netflix: the Origin of Chaos Engineering](#) - Learn the origins and history of Chaos Monkey and see why Netflix needed to create failure within th...
49. [Digital Twins vs. Traditional Load Testing Methods - Anvil Labs](#) - This proactive approach can help catch issues early, avoiding costly repairs or failures. Platforms ...
50. [Digital twins show great promise in civil engineering. But what's next?](#) - The twin captures the physical reality, and AI interprets it – identifying stress patterns, predicti...
51. [Quantum Computing Vs. Classical Computing - BlueQubit](#) - Learn the key differences between quantum vs classical computing, from qubits to exponential scaling...
52. [PDF Preparedness Wargaming for Critical Infrastructure Resilience: Taiwan ...](#)
53. [DARPA Funds Acoustic Warfare Payloads \\$9.6M - LinkedIn](#) - 🚨 DoD / DARPA Contract Alert for Subcontractors | News Anchor Brief 🚨 HEADLINE: \$9.6M DARPA Option F...
54. [\\$2M DARPA grant funds technology development for force-detecting ...](#) - The next generation of high-performance sensors for detecting force that can perform under extreme c...
55. [Post-quantum cryptography in 2026](#)

56. The Societal Implications of Quantum Computing on Security ... - This review will focus specifically on these societal implications, highlighting the dual nature of ...
57. Analysis: Quantum Medicine's Promise Raises New Privacy And ... - An analysis argues that quantum technology could accelerate drug discovery and improve diagnostics, ...
58. [PDF] Ethical Challenges of Dual Use Technologies - Preface. This 3 PDH course addresses the ethical challenges faced by engineering professionals when ...
59. [PDF] Exploring the use of quantum computers for resilience analysis in ...
60. Quantum-Ready Architecture for Security and Risk Management ...